

Bezawit Negash

(B.pharm, MSc. in Health Supply Chain Management)

ASSIGNMENT II

Report and Requisition Form (RRF)													
Name of Health Facility: Tikur Anbessa Specialized Hospital		Region: Addis Ababa											
		Maximum Stock Level = 4 Months of Stock											
Reporting Period:		Emergency Order Point = 0.5 Months of Stock											
Ser. No.	Product Description	Unit of Issue	Report Part						Requisition Part				
			Beginning Balance	Quantity Received	Loss/ Adj	Ending Balance		Calculated Consumption	Days Out of Stock	Maximum Stock Quantity	Quantity Needed to Reach Max	Quantity to Order	Remark on Quantity to be supplied (For PFSA Hubs Only)
						DU	Store						
A			B	C		D	E	F= A+B ±C - D- E	G	H=(120xF)/(60-G)	I=H-E-D	J	K
Acyc-32	Acyclovir - 500mg - Powder for injection	10				0		0	0	0	0		
Adre-30	Adrenaline (Epinephrine) - 0.1% in 1ml ampoule -Injection	10				0		0	0	0	0		
Allo-10	Allopurinol - 100mg - Tablet	10				0		0	0	0	0		
Amio-10	Amiodarone - 100mg - Tablet*	10				0		0	0	0	0		
Amox-21	Amoxicillin - 250mg/5ml - Oral Suspension	10				0		0	0	0	0		
Amcl-20	Amoxicillin + Clavulanic Acid - (125mg +31.25mg)/5ml - Oral Suspension	10				0		0	0	0	0		
Amcl-14	Amoxicillin + Clavulanic Acid - (500mg + 125mg) - Tablet (Film Coated)	10				0		0	0	0	0		
Ampi-31	Ampicillin Sodium - 500mg in Vial - Injection	10				0		0	0	0	0		
AntD-31	Anti-Rho (D) Immune Globulin - 300mcg in 2ml Vial- Injection	10				0		0	0	0	0		
Aten-11	Atenolol - 50mg - Tablet	10				0		0	0	0	0		
Azit-20	Azithromycin - 200mg/5ml - Oral Suspension	10				0		0	0	0	0		
Azit-10	Azithromycin - 250mg - Tablet	10				0		0	0	0	0		
Becp-41	Beclomethasone Propionate - 100mcg/dose - Aerosol (Oral Inhalation)	10				0		0	0	0	0		
Bisa-10	Bisacodyl - 5mg - Tablet	10				0		0	0	0	0		
Products with shelf life ≤ 6 months (S/No, Quantity and Expiry date): respectivilly													
Completed by : Group I		Signature:											
Verified by :		Signature:											
Approved by :		Signature:											
Distribution: First Copy to Supplying Hub, Second copy to RHB/ZHD/WoHO and Third Copy to remain with the pad													

<

>

Quarter RRF

Quarter II RRF

QuartellIRRF

Quarter IV RRF

+

⋮

◀

STOCK STATUS ANALYSIS

Stock Status Analysis

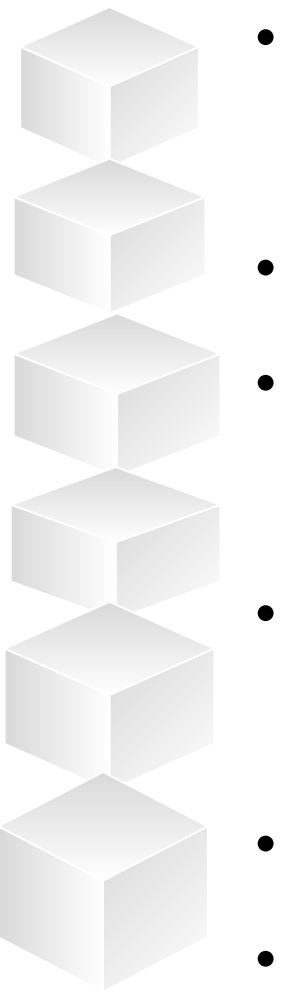
- Stock status analysis is a method to know how long medical products on hand will last.
- Any medical store manager should perform stock status analysis of available products to prevent stock-outs and reduce wastage and inventory holding cost.

How to Assess Stock Status

- A stock status analysis is done by determining the month of stock available in the medical store.
- The month of a stock determined as below: -

-

**Months
of stock
on hand**



Categorizing inventory medical products

ABC analysis

- ABC analysis is the process of dividing medical products of the medical store into **three classes** according to their **monetary value** of **consumption or usage** so that store managers or supply managers can focus on items that have the **highest monetary value..**

-

Cont . . .

- **Class “A” : The important few,**
 - 10- 20% of products, account for 70-80 % of the monetary value
- **Class “B” : The middle share**
 - 10-20% of products, 15-20% of the monetary value
- **Class “C” : the trivial many**
 - 60-80% of products, 5-10 % of the monetary value
- Class A medical products have more impact on the health facility budget and need strict control
-

Steps in performing ABC analysis:

- Step 1. List all items purchased and enter the unit cost.
- Step 2. Enter consumption quantities for each item.
- Step 3. Calculate the value of consumption for each item.
- Step 5. Calculate the percentage of total value represented by each item.
- Step 6. Calculate the cumulative percentage of total value for each item.
- Step 7. Choose cut-off points for A, B, and C
- Note: The results of ABC should be reconciled with that of VEN.

Custom sort the list based on annual values from largest to smallest

Pharmaceutical Name	Price	Annual Demand	Annual value \$	cum \$ value	Cumulative %	ABC classification
Nilotinib - Capsule - 200mg	24.99	334.00	8346.66	8346.66	37.97	A Items 80% of value 20% of SKU
Metoclopramide - Tablet - 10mg	7.70	675.00	5197.50	13544.16	61.62	
Enalapril Maleate - Tablet - 10mg	4.45	612.00	2723.40	16267.56	74.01	
Clindamycin - Injection - 150mg/ml	6.10	220.00	1342.00	17609.56	80.11	
Calcium Folate (Leucovorin Calcium) - Injection - 50mg/5ml in Vial	1.32	786.00	1037.52	18647.08	84.83	B: Items 15% of values 30% of SKUs
Prednisolone - Tablet - 5mg	2.25	260.00	585.00	19232.08	87.49	
Sulphamethoxazole + Trimethoprim - Injection - (400mg+80mg)/5ml in 10ml Ampoule	0.77	690.00	531.30	19763.38	89.91	
Diclofenac Sodium - Tablet - 50mg	6.22	66.00	410.52	20173.90	91.78	
Captopril - Tablet - 12.5mg	3.86	89.00	343.54	20517.44	93.34	C Items: 5% of value 50% of SKU
Acetylsalicylic Acid - Tablet (Enteric Coated) - 81mg	3.10	110.00	341.00	20858.44	94.89	
Dexamethasone - Injection - 4mg/ml	0.77	388.00	298.76	21157.20	96.25	
Norfloxacin - Tablet - 400mg	7.75	24.00	186.00	21343.20	97.10	
Paclitaxel - Injection - 100mg/16.7ml	12.80	14.00	179.20	21522.40	97.9129	
Methyldopa - Tablet - 250mg	0.85	148.00	125.80	21648.20	98.49	
Sodium Chloride (Normal saline) - Intravenous Infusion - 0.9% in 1000ml	2.85	43.00	122.55	21770.75	99.04	
Flumazenil - Injection - 0.1mg/ml in 5ml ampule	1.23	52	63.96	21834.71	99.33	

Assignment III

	A	B	C	D	E	F	G
1	Pharmaceutical Name	Price	Annual Demand	Annual value \$	cum \$ value	Cumulative %	
2	Acyclovir - 500mg - Powder for injection						
3	Adrenaline (Epinephrine) - 0.1% in 1ml ampoule - Injection						
4	Allopurinol - 100mg - Tablet						
5	Amiodarone - 100mg - Tablet*						
6	Amoxicillin - 250mg/5ml - Oral Suspension						
7	Amoxicillin + Clavulanic Acid - (125mg + 31.25mg)/5ml - Oral Suspension						
8	Amoxicillin + Clavulanic Acid - (500mg + 125mg) - Tablet (Film Coated)						
9	Ampicillin Sodium - 500mg in Vial - Injection						
10	Anti-Rho (D) Immune Globulin - 300mcg in 2ml Vial- Injection						
11	Atenolol - 50mg - Tablet						
12	Azithromycin - 200mg/5ml - Oral Suspension						
13	Azithromycin - 250mg - Tablet						
14	Beclomethasone Propionate - 100mcg/dose - Aerosol (Oral Inhalation)						
15	Bisacodyl - 5mg - Tablet						
16							
17							
18							
19							

Application of ABC analysis

- Measure the degree to which actual consumption reflects public health needs and morbidity.
- Reduce inventory levels and costs by arranging for more frequent purchase or delivery of smaller quantities of class A items
- Reduce inventory of items that have limited use, but cost the system large amounts of money
- Provide information for choosing the most cost-effective alternatives and finding opportunities for therapeutic substitution
- Gather information for pharmacoeconomic analysis.

FSN analysis

- FSN analysis is the process of classifying the health products based on their **movement from the inventory** for a specified period.
- This analysis helps the storekeeper and procurement department to keep the fast-moving items always available and take necessary measures towards the non-moving inventories.
-

Cont . . .

- FSN Analysis classifies inventory based on quantity, the rate of consumption and frequency of issues and uses.
 - F stands for Fast-moving,
 - S for Slow-moving and
 - N for Nonmoving items.

Cont . . .

- Inventory (material) turnover ratio (ITR): it indicates how many times the average inventory in stock will be converted into finished goods during specified period.
- $ITR =$
- $Products\ consumed = Beginning\ balance + Received - Closing\ balance$

Cont . . .

- High ITR- Efficient use of stock(F)
- Low ITR- Inefficient use of stock(S)
- 0 ITR- No use of Stock (N)

Cont . . .

-
- The following details are given to you, your hospital have closed the fiscal year and the transaction for three items are provided to you as below
-

• Particulars	• Atrovastatin - Tablet - 20mg	• Metoclopramide - Tablet - 10mg	• Paracetamol - Tablet - 500mg
• Stock as on 06-08-2018	• 830,000	• 120,000	• 140,000
• Stock as on 05-08-2019	• 513,200	• 24,500	• 30,000
• Purchase during the year	• 200,000	• 60,000	• 40,000

Cont . . .

- Atrovastatin –
 - Product consumed= $830,000+200,000-513,200=$ **516,800**
- Methoclopramide–
 - Product consumed= $120,000+60,000-24,500=$ **155,500**
- Paracetamol–
 - Product consumed= $140,000+40,000-30,000=$ **150,000**

Cont...

- Atrovastatin –
 - $= 515,000$
- Methoclopramide–
 - $= 72,250$
- Paracetamol–
 - $,000$

VEN analysis

- It is a method to set up priorities for medicines according to their health impact into **Vital (V)**, **Essential (E)**, and **Non-essential (N)** categories.
- It is a well-known method to help set priorities for purchasing medicines and keeping stock.
-

VEN analysis



V

- Potentially lifesaving
 - A patient may die or maybe harmed or disabled because of lack of this medicine
 - Crucial to provide basic health services (in its absence, service may be discontinued).
 - Must always be available
-



E

- Effective against less severe but significant illness, not vital;
 - It is lifesaving; without which patient may be in difficulty/ problem /maybe somehow substituted
 - Essential to the service without which it is difficult to give the health service
-



N

- Effective for minor illnesses and low therapeutic advantage
 - Important to patients; however, patients will not die because the absence of these medicines
 - Necessary to give the health service; however, health service delivery will not be discontinued in the absence of these medicines
-

Application of the inventory categorization techniques

- Generally, application of the inventory control techniques can be grouped under;

1. Inventory management policies

- Determining order frequencies: ordering class A items more often and in smaller quantities should lead to a reduction in inventory holding costs.

2. Procurement application

- Selection: VEN, ABC-VEN matrix analysis help managers identify purchases made for items that are not on the formulary or essential medicines list or not approved for use in the supply system

Cont . . .

- Procurement: ABC analysis will help in determining source of lower priced products, assuring procurement is in line with public health priorities

3. Warehouse application

- Arrangement of stock in warehouse - For proper product location in the warehouse/medical store e.g. FSN, ABC Combining two techniques
-

